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Inferring stochastic dynamics by biophysical Neural ODE using single-cell transcriptomics

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Abstract

Single-cell RNA sequencing has revolutionized our ability to dissect cellular heterogeneity and study cell fate mechanisms, yet inferring stochastic dynamics from static snapshots remains a fundamental challenge. Current approaches face a critical trade-off: mechanistic models impose rigid assumptions limiting biological realism, while data-driven methods sacrifice interpretability for deeper mechanistic explorations. Here, we present DynNet, a deep learning method that integrates Neural ODEs with biophysical models and prior knowledge of gene expression dynamics. DynNet learns the stochastic dynamics of gene regulatory systems for cell fate decisions. Benchmarking on synthetic data shows DynNet's ability to infer stable cell states, reconstruct dynamical trajectories, and characterize multi-stable cell fate transitions. Using hepatocyte differentiation data, DynNet demonstrates its capability to infer developmental trajectory and the underlying cell fate landscape, revealing the stability and transition probabilities among distinct cell states. Applied to Epithelial-mesenchymal transition (EMT) data, DynNet further captures critical gene regulations and transition paths during EMT.

Introduction

Stochastic dynamical transitions are fundamental in biological, physical, and chemical systems^{1–5} and are especially crucial for understanding cell fate determinations^{6,7}. The existence of multiple stable steady states allows cells to transition between distinct fates in response to external or internal cues. This mechanism is particularly important in stem cell differentiation⁸, immune cell activation⁹, and cancer cell phenotypic switching^{10,11}.

Single-cell RNA sequencing (scRNA-seq) technology has enabled the study of cellular functions^{12,13}, including lineage decision, gene regulatory networks, and disease mechanisms at cellular resolution^{14–16}. However, scRNA-seq involves the destruction of cells to measure their molecular state, making it impossible to directly track the specific molecular trajectories of individual cells over time¹⁷. This limitation in capturing temporal dynamics has spurred the development of computational approaches to reconstruct the dynamics of system behaviours.

Various computational methods have been developed, including Markov-model-based approaches and deep learning methods, to reconstruct cell fate trajectories and model gene expression dynamics^{18–22}. These methods are now widely used to reconstruct developmental processes^{22–26} and disease progression^{27,28}. Markov-model-based methods focus on reconstructing cell state trajectories and ordering cells in pseudo-time. Nevertheless, these types of methods lack a quantitative, interpretable dynamic model explicitly linked to underlying gene regulatory mechanisms^{18,19,29}. Neural ordinary equations (Neural ODEs) use neural networks and differential equations, allowing reconstruction of gene expression dynamics continuously from data^{20,21,30–32}. However, classic deterministic Neural ODEs fail to capture the multi-stable dynamics as only a definite direction of dynamics is allowed at any given point^{20,33–35}. Moreover, it is difficult for Neural ODE frameworks to include prior knowledge, for example, specific gene regulatory interactions, and often the dynamics on low-dimensional latent spaces is derived from the high-dimensional gene expression data, limiting the direct interpretability of the inferred dynamics at the

level of individual genes. RNA velocity-based methods like scVelo are effective for inferring cellular dynamics along trajectories by capturing the rates of spliced and unspliced transcripts^{16,24,31,36}. Multi-omics data, such as metabolic markers or lineage information, might be used to estimate unspliced RNA sequences³⁷. In the absence of additional measurements, inferring interpretable dynamics involving underlying gene regulatory mechanisms solely from scRNA-seq data remains a challenge. As gene regulatory network (GRN) models provide interpretable frameworks for understanding transition dynamics between different stable states^{38–42}, it's important to bridge mechanistic modelling and data driven approaches^{43–47}. In addition, energy landscape methods provide effective ways to characterize stability, transitions, and stochastic dynamics for cell fate decision processes in gene regulatory networks^{41,44,48}.

Here we present DynNet (Stochastic Dynamics Inference Network) to infer the latent gene regulatory dynamics. Crucially, DynNet leverages scRNA-seq data collected at multiple, discrete, and known time points, utilizing the temporal information as constraints to model the underlying continuous process. Our core idea is to employ stochastic differential equations (SDEs), the natural mathematical framework for stochastic simulation, to mechanistically decouple the dynamics into a deterministic gene regulatory drift term and a diffusion term representing stochastic fluctuations, allowing for interpretable, single-cell level inference. In this approach, the model jointly optimises biophysical parameters and network weights by incorporating prior regulatory knowledge via evidence-tiered constraints. To capture the inherent stochasticity of gene expression dynamics, DynNet employs a neural network to estimate the noise variance dynamically, allowing for a more accurate description of single-cell data. DynNet also integrates landscape reconstruction within its framework to infer the underlying dynamical structure of gene regulatory systems. By embedding energy landscape analysis directly into its modelling process, DynNet provides a tool to identify transition driver genes and important regulations, through perturbation analysis. To evaluate DynNet, we apply it to representative synthetic datasets, including mutually inhibited self-activation (MISA) model and cell differentiation gene network. DynNet can efficiently reconstruct steady

state distributions, cell state evolution vector field, and cell fate landscape, compared with ground truth data. We further apply DynNet to realistic biological datasets: the epithelial-to-mesenchymal transition (EMT) process with external signalling modulation⁴² and the hepatocyte differentiation system exhibiting bifurcation behaviour⁴⁹. Through DynNet, we reconstruct the developmental trajectory and underlying dynamic landscape for the hepatocyte differentiation. We also quantify the most probable state transition paths and identify key regulators driving EMT.

Results

The overview of the DynNet

We developed a hybrid modelling framework, DynNet (Figure 1), based on dynamical systems theory to reconstruct continuous cell fate transition dynamics. DynNet is based on discrete snapshot data in scRNA-seq across multiple time points. We define the cell data as being measured at a total of T time points, $t = 1, 2, 3, \dots, T$. For any given time point t , the number of cells is N_t , and the number of genes is M . We assume that the initial time point ($t = 1$) represents the starting point of the biological process, intermediate time points capture the transition states, and the final time point ($t = T$) corresponds to the terminal phenotype, such as fully differentiated cells or stable states. First, by pre-processing the raw data (Figure 1A), we transform the sequencing data of N_t cells at each time point into a gene expression matrix $\mathbf{X}_t \in \mathbb{R}^{N_t \times M} = \{\mathbf{x}_i\}$, where $\mathbf{x}_i \in \mathbb{R}^M$ represents a feature vector characterizing the state of the i -th cell. The core of this work is to model the unobservable transitions between time points based on dynamical systems theory. SDEs provide a natural formal framework for describing stochastic transitions between cell states^{50,51}. Therefore, we model gene expression dynamics using SDEs

$$d\mathbf{x} = \mathbf{f}(\mathbf{x})dt + \boldsymbol{\sigma}(\mathbf{x})d\mathbf{W} \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^M$ represents the state of gene expression at a certain time. The term $\mathbf{f}(\mathbf{x}) \in \mathbb{R}^M$ describes non-linear gene regulatory interactions, while $\boldsymbol{\sigma}(\mathbf{x})$ accounts for both biochemical noise (intrinsic

fluctuations) and environmental noise (extrinsic fluctuations). The noise $\mathbf{W}(t)$ represents the standard Brownian motion, with independent components satisfying $\mathbb{E}[\mathbf{W}_i(t)] = 0$ and $\mathbb{E}[\mathbf{W}_i^2(t)] = t$. According to SDE theory⁵², the noise term satisfies $\boldsymbol{\sigma}(\mathbf{x})\boldsymbol{\sigma}(\mathbf{x})^T = 2\mathbf{D}(\mathbf{x})$, where $\mathbf{D}(\mathbf{x})$ represents the diffusion coefficient matrix, and ε denotes the noise intensity. Under long-term equilibrium conditions ($t \rightarrow \infty$), the system dynamics converge to a steady state probability distribution P_{ss} . We employ a velocity estimation network to estimate $\mathbf{f}(\mathbf{x})$ and a noise estimation network to estimate $\mathbf{D}(\mathbf{x})$. To incorporate prior biological knowledge, the nonlinear function $\mathbf{f}(\mathbf{x})$ explicitly encodes gene regulatory relationships via a neural network layer embedding Hill functions (Figure 1C), with dynamics of gene i governed by

$$\mathbf{f}_i(\mathbf{x}) = \sum_{N_{ij}=1} \frac{a_{ij}x_j^{n_h}}{s_j^{n_h} + x_j^{n_h}} + \sum_{N_{ij}=-1} \frac{b_{ij}s_j^{n_h}}{s_j^{n_h} + x_j^{n_h}} - k_i x_i, \quad i = 1, 2, \dots, M. \quad (2)$$

where a_{ij} and b_{ij} represent activation and inhibition strengths from gene j to gene i , respectively, s_j denotes the threshold level corresponding to gene j , and k_i reflects the degradation rate of gene i . The model is constrained using a prior gene regulatory network (GRN) adjacency matrix N , where $N_{ij} = 1$ indicates activation, and $N_{ij} = -1$ indicates inhibition ($i, j = 1, 2, \dots, M$) (Figure 1B). n_h is the Hill coefficient, which was chosen to be 4 in DynNet based on the results of previous works^{53,54} (see Supplementary Information Section A.2 and Figure S1).

To ensure that the learned dynamics align with known biological network structures and to prevent overfitting sparse time-sampled data, we introduce two constraint mechanisms. Hard constraints mask neural connections that do not exist in the GRN, restricting parameter updates only to existing edges. Soft constraints penalize spurious regulatory effects via regularization (Methods). For the stochastic fluctuations in the SDE system, we employ an additional noise estimation network. The diffusion coefficient is estimated by an independent neural network, $\mathbf{D}_\theta$ (Figure 1C, right). This network

constructs the diffusion matrix, which is constrained to be a diagonal matrix to model biochemical fluctuations.

The training process follows an alternating optimization strategy (Figure 1E). First, the regulatory parameters of the velocity estimator are set as $\theta = (\mathbf{A}, \mathbf{B}, \mathbf{S})$. Here, $\mathbf{A}$ is activation interaction matrix, with its element a_{ij} representing the activation strength from gene j to gene i . $\mathbf{B}$ is inhibition interaction matrix, with its element b_{ij} , representing the inhibition strength from gene j to gene i . $\mathbf{S}$ is threshold vector, with its component S_j denoting the half-maximal effective concentration (threshold level) corresponding to gene j . These parameters are optimised by minimizing a composite loss function $\mathcal{L}_v$. This composite loss consists of a dynamical consistency loss $\mathbb{E}_T[|\mathbf{f}(\mathbf{x}_T)|^2]$, a weight-matching loss $\frac{1}{T} \sum_{t=1}^T \mathbb{E}_t[|\widehat{\mathbf{w}}_t - \mathbf{w}_t|^2]$, and a constraint term $\mathcal{C}(\theta)$. The expectation $\mathbb{E}_t[\cdot]$ in the loss definition is computed as the empirical average over the samples at time t . The dynamical consistency loss minimizes the drift term when the system is close to stable state, ensuring that the learned dynamics meet the stable state conditions. To guarantee consistent distribution weights among multiple stable states throughout the developmental process, we introduced a weight-matching loss (Figure 1D). This term minimizes the discrepancy between the empirical weights $\mathbf{w}$, which represents the distribution weights estimated from the data using a Gaussian Mixture Model (GMM), and $\widehat{\mathbf{w}}$, which is estimated from simulated data generated by the DynNet model. We provide the theoretical derivation and verify the stability of our designed loss function on multi-stable datasets (see Supplementary Information Section A.4 and Figure S2-S4). The constraint term imposes restrictions on the parameters in accordance with the prior GRN, thereby preventing overfitting to sparsely sampled temporal data (see Methods). The overall loss function is formulated as

$$\mathcal{L}_v = \mu_1 \mathbb{E}_T[|\mathbf{f}(\mathbf{x}_{t_n})|^2] + \mu_2 \frac{1}{T} \sum_{t=1}^T \mathbb{E}_t[|\widehat{\mathbf{w}}_t - \mathbf{w}_t|^2] + \lambda_1 \mathcal{C}(\theta), \quad (3)$$

where different hyperparameters combinations (μ_1 , μ_2 and λ_1 are determined in pre-experiments, see Supplementary Information Section A.5) are used to alternately train the parameter matrix $\mathbf{A}$ and $\mathbf{B}$, as well as the parameter vector $\mathbf{S}$, until convergence, serving as a description of the deterministic dynamics (Figure S2). After optimizing the velocity estimator parameters, the noise parameters ϑ are calibrated by minimizing a Kullback-Leibler (KL) divergence loss, which further refines the model's representation of both biochemical fluctuations and environmental noise. The loss for noise estimator is defined as

$$\mathcal{L}_{\mathcal{D}} = \frac{1}{T} \sum_{t=1}^T \mathbb{E}_t [\text{KL}(\hat{p}_{j,t} | p_{j,t})] + \lambda_2 \mathcal{R}_2(\vartheta) \quad (4)$$

where $\mathcal{R}_2(\vartheta)$ is a regularization term.

The stable state probability distribution $P_{ss}(\mathbf{x})$ is derived using moment equations and subsequently the energy landscape is calculated as $U(\mathbf{x}) = -\ln P_{ss}(\mathbf{x})$ (Figure 1F), which reveals the attractor structure and cell state stability^{48,54}. In this landscape, local minima correspond to stable cellular phenotypes, while saddle points indicate the critical points of fate transitions. The landscape not only elucidates the global stability of the attractors but also estimates the transition pathways between them. In summary, DynNet reconstructs continuous dynamical trajectories from discrete scRNA-seq data by integrating biophysical modelling with deep learning techniques. Its output includes reconstructing cell fate transitions, constructing cell type transition networks, analysing gene regulatory relationships, and predicting cell fates (Figure 1G). In the following sections, we demonstrate the effectiveness of DynNet by applying it to a variety of synthetic and realistic scRNA-seq datasets.

DynNet accurately recovers multi-stable dynamics in simulation data of MISA model

The Mutual Inhibition and Self-Activation (MISA) model^{54,55} is a classical gene regulatory motif in which two genes mutually inhibit each other while self-activating themselves (Figure 2A and Figure S5). The system's multi-stability property makes it a mathematically tractable benchmark for dynamic inference methods. We generated synthetic data comprising five temporal snapshots (2000 samples per time point), capturing transitions from initial states to equilibrium. Phase space analysis of the simulated data revealed three stable states: an X1-dominant differentiated state (S1), an X2-dominant differentiated state (S2), and a balanced pluripotent state (S3) with moderate expression levels (Figure 2B). To evaluate DynNet's performance in multi-stable systems, we established a comprehensive analysis framework assessing three key aspects: stable state localization accuracy, single-cell fate prediction reliability, and global dynamic reconstruction capability. This framework allows for a comprehensive verification of the model's capacity to resolve multi-stable configurations, track single-cell developmental paths, and recover dynamical landscapes.

To assess the ability of DynNet to recover and quantify the properties of the system's stable states, we compared its predictions against the ground truth simulation data of MISA. DynNet correctly identified the presence of three stable states at the last time point. We then quantitatively compared DynNet's estimated positions and variances of these stable states to those derived from ground truth data. This comparison indicates that DynNet accurately estimates stable state centers, showing close agreement with observed data (Figure 2C). Furthermore, the covariance regions, reflecting stable state variability, also show alignment with the simulation data. Beyond position and covariance, DynNet's estimates of the marginal distributions for variables X1 and X2 also closely match those observed in the ground truth data (Figure 2D and Figure S6). The reconstructed energy landscape shows agreement with the overall distribution and shape of the data (Figure 2E). Consistent performance in reconstructing stable state properties was also observed when applying DynNet to bistable data (Figure S7 and S8). Collectively, these findings demonstrate DynNet's capacity to accurately identify and quantify key stable state properties, including their number, locations, associated uncertainties and marginal distributions. This

demonstrated capacity is crucial for precisely defining distinct cellular states and confirms DynNet's reliability in accurately predicting and comprehensively characterizing the stable states of multi-stable systems.

To validate single-cell fate predictability, we assessed DynNet's performance by formulating stable state prediction as a classification task based on decision boundaries. As a quantitative measure of prediction reliability, decision boundaries delineate the regions of phase space leading to different stable states (see Supplementary Information Section D). The grid-structured dataset produced by the MISA model delineates the ground truth decision boundaries, serving as a benchmark for our comparative analysis (Figure 2F left). We then applied the trained DynNet model to predict the stable state for each point in the grid-structured dataset and derived its corresponding decision boundaries (Figure 2F right). DynNet's predicted decision boundaries demonstrate consistency with those obtained from the simulation data, accurately dividing the phase space into the three stable state basins of attraction. This agreement further demonstrates the effectiveness of DynNet in delineating the region of attraction in phase space. We also compare the decision boundaries between DynNet and Neural ODE method as benchmark method (see Supplementary Information Section C). The decision boundaries predicted by DynNet are more accurate than those predicted by the classic Neural ODE method. These results demonstrate that for some cells with an undetermined fate, DynNet can effectively predict their future fate (Figure S9 and S10).

To evaluate DynNet's ability to reconstruct the system's global dynamic behaviour, we first assessed its performance in reconstructing the overall velocity field. We computed the velocity field of the simulated data as a benchmark (Figure 2H and Figure S11). The velocity field estimated from DynNet not only captured key features such as decision boundaries and stable states but also demonstrated better performance in approximating the true dynamics compared to the Neural ODE method across datasets with varying numbers of time points (2, 3, and 5) (Figure 2G and Figure S11). Unlike the traditional Neural ODE, which struggled to accurately reconstruct dynamics with fewer time points and required more data for comparable accuracy (Figure S11), DynNet consistently provided more accurate predictions

across all time points tested. These results underscore the robustness of DynNet in reconstructing the underlying dynamics of the system with limited temporal data. Beyond reconstructing the overall velocity field, we also evaluated DynNet's ability to capture dynamic transitions between stable states. To specifically validate this ability, we used a perturbed bistable system. Data was simulated from this system by modifying parameters to alter stable state weights without changing their positions, thereby capturing a clear transition from the low- X_1 /high- X_2 state to the high- X_1 /low- X_2 state (Figure S7 and S12). DynNet accurately identified this transition when inferring dynamics from the simulated data (Figure S12). These results support that DynNet can robustly reconstruct the global dynamics and state transitions of multi-stable systems.

To validate DynNet's reliability in practical applications, we evaluated its robustness against violations of modeling assumptions and its optimization stability. First, regarding the robustness to noise structures, we verified that DynNet maintains high fidelity even when the underlying data exhibits non-diagonal covariance (Figure S13). Notably, explicitly modeling the full non-diagonal diffusion matrix yielded negligible performance gains on these simulated datasets. These results indicate that the diagonal diffusion assumption provides a robust approximation, effectively balancing computational efficiency with modeling fidelity. To further bridge the gap between continuous SDE modeling and the discrete, sparse nature of scRNA-seq data, we validated DynNet on Gillespie-simulated trajectories with binomial dropout (Figure S14). The model successfully recovered the sparse regulatory topology (with near-zero off-diagonal terms) and accurately quantified cell fate biases, demonstrating robustness to realistic measurement noise. We further evaluated the stability and efficiency of the optimization process, finding that the model exhibits high parametric consistency by converging to the same set of regulatory parameters across independent random initializations (Figure S15). Additionally, a comparison of initialization strategies reveals that incorporating the GRN prior significantly accelerates loss convergence compared to random initialization, empirically validating the computational necessity of our prior-guided training strategy (Figure S16).

DynNet accurately recovers simulation data for developmental branching process

To further evaluate DynNet's performance, we constructed a gene regulatory network comprising 7 genes (Figure 3A and Figure S17). In this network, red arrows denote activation, while blue arrows indicate repression. Here, G1, G2 and G3 progressively activate G4 and G5, which mutually inhibit each other, forming a toggle switch, and G6 and G7 are downstream targets of G4. BoolODE, a stochastic differential equation-based simulator was used to simulate the multi-stable dynamics⁵⁶. Training data comprise six temporal snapshots capturing the transition from the initial state to equilibrium, which are reduced in dimensionality using Principal Component Analysis (PCA) (Figure 3B). This system exhibits differentiation-like dynamics, as the system state evolves from a stem-cell-like state (I) into two distinct differentiation states (D1 and D2) over time. Comparing the gene expression in the two differentiation states, we found that genes G4, G5, G6, and G7 showed significant differences between the two stable states, which is consistent with the gene regulatory relationships (Figure 3C).

DynNet successfully reconstructed the system's dynamics from simulated data. The marginal distributions of the estimated stable states demonstrate agreement between DynNet's estimates and the data (Figure 3D). The estimated velocity field accurately captures gene expression trajectories toward stable states (Figure 3E). However, we also noted that the estimated velocity fields do not fully align with the cell trajectories before cell fate bifurcation. This apparent misalignment may arise because the overlap between pre-bifurcation states and the bifurcation points in the projection space (Figure 3B) makes it challenging to accurately assess the dynamics in this visualization. Despite using data generated by a simulator based on a different dynamic modelling framework from being used by DynNet, the model successfully estimated the dynamics. Using the gene expression dynamics inferred by DynNet, we analysed the temporal changes in individual genes. We specifically observed the cascade activation dynamics involving gene G1, G2, and G3 (Figure S18), which showed characteristic transient expression profiles with levels first increasing and then decreasing over time. Among the downstream targets, we focused on G6 and G7, which serve as marker genes distinguishing different cell fates. The mutual

inhibition between G4 and G5, acting as cell fate switches, then causes a significant and synchronous bifurcation in the expression levels of both G6 and G7, clearly marking the system's bistable fate choice (Figure 3F). Consistent with the MISA validation, DynNet accurately recovered the bifurcation dynamics and steady-state distributions from Gillespie simulation data (Figure S19), further confirming its robustness to realistic molecular fluctuations. We also validated the performance of DynNet on this dataset with an extra non-diagonal noise as environment noise (Figure S20 and S21). Despite being trained solely with the diagonal diffusion matrix, we confirmed the DynNet's ability to reconstruct the attractors with accurate noise estimation. Furthermore, quantitative accuracy showed minimal relative error in inferred stable state weights and agreement in the marginal density plots (Figure S22). These results support our modeling choice with the diagonal diffusion matrix assumption.

We next investigated the impact of prior knowledge constraints on model inference. Using hard constraints where the regulatory relationships in the simulation data were predefined, the interaction type between two genes remains fixed during optimization, and crucially, parameters for non-existent interactions (not present in the prior) are rigidly masked or fixed to zero. For example, if the prior dictates that gene j has no inhibitory effect on gene i , the corresponding inhibitory parameter is set to zero and is not optimized. In contrast, soft constraints do not rigidly mask parameters but instead penalize them during optimization, driving them to be as small as possible (Figure S23 and S24). Figure 3H illustrates the training loss trajectories, and this flexibility comes at a cost: soft constraints lead to slower convergence compared to hard constraints due to the expanded parameter search space. Despite the computational cost, this difference allows soft constraints to explore the parameter space and potentially discover new interactions. However, we noted that when using soft constraints, the estimated parameters occasionally diverged from the true underlying network (Figure 3G and Figure S24). The edges not present in the original network yet still show nonzero estimated values after optimization (Figure 3G), such as the activating edge from G5 to G7. This phenomenon is biologically explicable by the indirect interaction in the true network: G5 inhibits G4, and G4 in turn inhibits G7. This double-inhibition induces

a positive correlation between G5 and G7 expression, which the model interprets as a direct regulatory edge under soft constraints. Nonetheless, for edges present in the prior network, DynNet produced reasonable weight predictions. While soft constraints require careful interpretation of such indirect relationships, they offer the significant advantage of allowing the model to infer potential regulatory relationships, a capability not possible with rigid hard constraints. Notably, this implies that the soft constraint approach may be viewed primarily as a strategy to enhance the reconstruction of system dynamics by capturing functional dependencies, rather than as a dedicated tool for precise structural network inference. Building upon this capacity to capture functional dynamics, DynNet naturally supports *in silico* perturbations. Simulated suppression of the driver gene (G4) in the 7-gene network correctly predicted a macroscopic shift from a developmental bifurcation to a monostable trajectory (Figure S40). To investigate this discovery capability when prior knowledge is incomplete, we performed experiments on the 7-gene network with only self-activation terms as prior, leaving all cross-gene regulatory interactions to be inferred. In this case, hard constraint did not lead to the desirable dynamics because of the loss of prior information. But DynNet successfully reconstructed the underlying dynamics with a possible GRN structure based on the dynamic data under soft constraint (Figure S25). Furthermore, we demonstrate that DynNet can learn dynamics through a two-step method of topology-hard constraint inference parameters in a soft constraint learning network (see Supplementary Information Section E.2 and the Figure S36-38). To validate this structural identifiability and the utility of data-driven priors, we evaluated DynNet across 50 synthetic networks (see Supplementary Information Section E.4). Initialization with a correlation network acted as an optimization accelerator, speeding up convergence while enabling the soft-constraint to robustly correct false prior edges (Figure S46). Consequently, DynNet demonstrated reliable *de novo* topological recovery, maintaining robust accuracy under purely data-driven settings (correlation network as prior) and even when guided by a 30% corrupted prior (Figure S39).

Given these advantages, we further tested DynNet's inference capability under limited observational data. We designed a 9-dimensional gene network and considered the scenario where only a subset of genes could be observed and performed dynamical estimation based on this limited information (Figure S26). The simulated data evolved into three discrete stable states, each corresponding to distinct cell types characterized by the expression of genes G, H, and I. We selected a subset of genes for training DynNet to fit the multi-stability data generated by the model and applied soft constraints for regularization. DynNet's estimated stable states show alignment with the simulation data in terms of their centers, marginal distributions, and covariance regions. Although some deviations were observed between the estimated results and the data, particularly in the marginal distributions of genes E and F where two low-expression states were less distinctly separated in the simulation data compared to the estimates, the overall consistency and biological interpretability remained evident (Figure S26). The test results on the synthetic gene network demonstrate that DynNet effectively handles complex gene regulatory dynamics of cell differentiation, even when data is incomplete, especially through indirect regulation (Figure S27). This capability to reconstruct dynamics from incomplete node observations is particularly crucial given DynNet's fundamental design philosophy. Rather than attempting to model the entire transcriptome, DynNet focuses on capturing the continuous dynamics of a targeted set of core driver genes governing cell fate decisions. Because the complete biological network inevitably involves unobserved intermediate regulators, the ability to effectively absorb these hidden effects into the core network via inferred indirect interactions is paramount. By doing so, DynNet ensures the recovery of macroscopic system dynamics even when the analysis is strictly confined to a limited core gene set.

We also investigated the impact of sparse temporal sampling by determining the required temporal resolution on synthetic datasets. We evaluated the impact of sparse temporal sampling across multiple dimensions. While sparse sampling yielded high error and variance, prediction accuracy, loss convergence, and structural recovery all optimally plateaued between 4 and 6 time points (Figures S41-S44). This confirms that a practical resolution of 4 to 6 time points, which aligns well with standard

scRNA-seq experimental designs, provides sufficient temporal anchors for robust continuous dynamic inference.

DynNet deciphers bifurcation dynamics and fate determination in hepatocyte differentiation

To assess DynNet's capability in inferring stochastic dynamics in real biological systems, we applied DynNet to a bifurcating dataset from hepatocyte differentiation. In vitro differentiation of pluripotent stem cells bifurcates into distinct cell fates, such as hepatocyte-like cells or liver bud (LB) formation, depending on the culture conditions⁴⁹. Recent studies have systematically analyzed the differentiation trajectory from induced pluripotent stem cells (iPSCs) to mature hepatocyte-like (MH) state and LB state. During early differentiation, iPSCs first transition to the definitive endoderm (DE) stage before progressing to the hepatic endoderm (HE) state. At this stage, distinct culture conditions drive two divergent pathways: conventional conditions yield immature hepatoblast-like (IH) cells that mature into hepatocyte-like cells, whereas organoid cultures induce the formation of LB tissues (Figure 4A). Here, we evaluate the feasibility of reconstructing this bifurcation process using DynNet.

For simplicity, we identified 20 candidate key genes for further investigation through differential gene expression analysis and functional annotation (Figure S28). We computed the asynchronous correlation among candidate genes by matching data across different time points through data-driven method (see Supplementary Information Section F.1). This step ensures that gene regulatory relationships are captured while accounting for temporal dynamics. Based on the calculated asynchronous correlations, we estimated a gene regulatory network from the candidate genes. The estimated gene-gene interactions were visualized using a circular plot, with red edges indicating activation and blue edges indicating inhibition. This network serves as the reference framework for DynNet analysis (Figure 4B and Figure S29). Dimensionality reduction of the key genes revealed the underlying developmental trajectory, where the

first principal component primarily captured differentiation progression, and the second one reflected cellular heterogeneity. Notably, the two terminal cell states exhibited clear separation (Figure 4C).

To analyze this bifurcation dataset, we employed DynNet to characterize its dynamic properties and regulatory interactions. The dynamic modelling allowed us to validate the branching trajectories and molecular mechanisms that regulate hepatocyte differentiation. Figure 4D presents a comparison between the expression density distributions of several genes in the estimated stable states and the corresponding empirical data. This comparison demonstrates that DynNet accurately estimates the marginal expression profiles observed in both terminal differentiation states. Concurrently, we computed the velocity field and visualized it on the energy landscape to reveal the global dynamic properties of the system. The system showed two stable attractors corresponding to MH and LB states, with velocity streamlines converging towards these attractors. Additionally, the velocity fields at the HE cluster exhibits bifurcation dynamics, consistent with experimental observations of HE cells displaying different differentiation trajectories under varying culture conditions (Figure 4E). To investigate developmental transition pathways, we quantified transition actions between different cell states and estimated their associated probabilities⁵⁷. The results show that intermediate states lowered transition actions, for example, the $DE \rightarrow HE \rightarrow \text{early LB}$ path required less action than a direct $DE \rightarrow LB$ transition (Figure 4F). This is consistent with our previous findings showing that intermediate states can accelerate cell fate transitions⁴². Reverse transitions consistently exhibit higher action costs than forward developmental transitions. Additionally, transitions to MH states require less action than those to LB states, suggesting a developmental bias toward MH. First passage time (FPT) analysis⁵⁸ further support this finding, showing that MH states have shorter FPTs than LB states, aligning with their lower transition action costs. To assess robustness, we evaluated DynNet across different simulated timescales and found that FPT estimates remain consistent across stable states, suggesting that DynNet effectively captures the system's global dynamical features (Figure 4G).

Finally, we performed *in silico* knockout experiments in DynNet's dynamic model. We targeted highly expressed genes in distinct cell states to assess their impact on differentiation. Knocking out these lineage-specific markers redirected differentiation, highlighting their role in cell fate determination (Figure 4H). Eliminating MH-specific genes such as *TOB1*, which maintain mature hepatocyte function, accelerated the transition toward LB states, indicating their critical role in preserving mature cell identity, consistent with existing biological evidence^{59,60}. In contrast, removing LB markers such as *RANBP1* promoted differentiation toward the MH state. This result suggests that these genes play a crucial role in regulating the balance between progenitor and differentiated states, consistent with previous findings on lineage commitment in hepatocyte development⁶¹. Similarly, knockout of stem cell markers such as *SRGAP2* did not affect the final outcome but altered the speed of differentiation, likely due to the reduced stemness of the cells. This observation aligns with studies indicating that stem cell markers regulate pluripotency and proliferation rather than directly determining cell fate⁶². First passage time (FPT) analyses further support these findings: knocking out MH markers led to significantly reduced LB FPTs, reflecting a faster transition, and vice versa (Figure 4I and Figure S30). For genes whose mechanisms are not yet well understood, DynNet can predict their impact on the differentiation process through knockout experiments, as demonstrated with *CHP1*, *SRGAP2* and *SRSF9*. Overall, the knockout simulations provide valuable insights into how intercellular interactions and gene regulatory mechanisms drive cell fate decisions.

More broadly, our findings demonstrate that DynNet enables a detailed reconstruction of hepatocyte differentiation dynamics at the single-cell level. The enhanced resolution provided by DynNet allows us to capture the evolving dynamics and regulatory interactions, revealing the gradual transitions between distinct cell fates. Although cell fate bifurcation is a complex process, DynNet reliably recapitulates both the stable state distributions and dynamics observed in experimental data. These results underscore DynNet's potential for revealing the mechanistic underpinnings of cellular differentiation and fate determination.

DynNet decodes EMT landscape switching

To see the effectiveness of DynNet in different biological data, we applied it to a TGF- β perturbation lung cancer time-course dataset for EMT. We analyzed dynamics of this single-cell dataset measuring 36 genes at seven time points (Day 0, 2, 6, 10, 12, 16, and 20) in over 100,000 cells. TGF- β was applied for the first 10 days to induce EMT and then withdrawn for the next 10 days to promote the mesenchymal-to-epithelial transition (MET), a reversal in cell fate. We estimated the data-driven energy landscape from the real data, representing it in the projection space (Figure 5A, Figure S31-S33 and Methods) and performed K-means clustering to identify four clusters, corresponding to terminal states E, M, IM (MET), and pEMT (partial EMT state), in agreement with previous work⁶³. First, we focused on the period from Day 10 to Day 20, when cell fate transitions are driven solely by gene regulations.

The MET process begins at Day 10 and stabilizes by Day 20. To investigate these dynamics, we first calculated the data-driven landscape, estimated from the distribution of scRNA-seq data (Methods). This landscape was visualized by projecting the high-dimensional data onto a one-dimensional axis using PCA (Figure 5B). The position along this axis represents the progression through the MET process, and the height of the potential landscape at each point indicates the potential (or energy) of the corresponding cell state. At Day 10, the M state exhibits low energy, reflecting its stabilization following the TGF- β -induced transition from the E state (Figure 5B, Top). After TGF- β removal, the M state's energy increases and the E state's decreases (Figure 5B, Bottom). These shifts suggest that the energy landscape underlies the transition from the M state to the E state. The global stability of this system converges to the E state.

To simplify DynNet's parameters and reconstruct the MET dynamics, we focused on a subset of key genes driving the cell fate transition. As DynNet can capture indirect regulatory interactions, reducing the gene to a subset should still reveal the main dynamic trajectories. We performed PCA using PC_E and PC_M , derived from E and M markers respectively, and calculated each gene's contribution along these axes (Figure S33). This analysis revealed that only a few genes distinguish the E and M states. Based on

the bimodality coefficient of gene expression from Day 10 to Day 20, we selected five core genes (Figure S33 and S34). For the selected core genes, we investigated currently known potential gene interactions and constructed a prior network (see Supplementary Information Section F.2). Since the regulatory relationships among genes are not fully understood, we applied soft constraints as prior information.

We trained DynNet on the MET data to estimate the dynamics governing cell fate transitions. Figure 5C displays the model-based energy landscape, derived from DynNet's learned dynamics, with gene expression data at Day 20 overlaid. The alignment of low-energy regions with high-probability areas indicates that DynNet accurately captures the system's dynamic behaviour. Stable state simulations reveal that the potential distributions of E and M markers clearly distinguish the two states, a finding further supported by the marginal gene expression distributions (Figure 5D), which show that DynNet effectively estimates both the width of gene expression peaks and the intrinsic noise levels. In addition, cells in the E state exhibit high E marker expression, while cells in the M state show high M marker expression, and intermediate cells express both (Figure 5E). The velocity field calculated in the projected space (Figure 5F) confirms the stability of the E and M states. However, not all cells revert to the E state, suggesting that the M state retains some stability. Furthermore, we estimated a weighted gene regulatory network based on the prior gene network (Figure 5G). The estimated results align well with biological evidence, revealing mutual activation among characteristic genes within the same cell type, while characteristic genes of different cell types predominantly exhibit inhibitory relationships.

Identify transition paths and critical genes for EMT

As DynNet reconstructs the underlying dynamics of EMT, we performed several downstream analyses. Based on the stable state positions on the landscape, the M-to-E transition likely occurs via the intermediate states IM and pEMT, consistent with previous findings⁶³. To verify this, we applied the minimum action path method based on Freidlin-Wentzell theory⁵⁷ to compute the transition actions (Figure 6A and B). Without TGF- β (Day 10-20), the transition action for MET is significantly lower than

that for the reverse E-to-M transition, indicating that MET is more probable under these conditions. Moreover, Figure 6A shows that EMT follows a direct path, while MET passes through IM, which corresponds to the lowest energy barrier and ensures system stability. Landscape analysis confirms a lower total action for MET, reinforcing the tendency toward the E state. Additionally, the lower action from IM-to-M compared to M-to-IM suggests that IM is a metastable state, with cells often reverting to the more stable M state. Finally, transition network analysis reveals that indirect paths have higher probabilities and lower actions than direct ones, further supporting the role of intermediate states in MET (Figure 6C and Supplementary Information Section F.2).

To further explore the effects of different genes on the system's dynamics, we performed a sensitivity analysis of DynNet's regulatory parameters. This analysis computes the changes in transition actions, $\Delta S_{E \rightarrow M}$ and $\Delta S_{M \rightarrow E}$ after individually perturbing model parameters. Specifically, we perturbed activation constants within groups and inhibition constants between groups by $\pm 5\%$ (Figure 6F and Supplementary Information Section F.2). The results indicate that enhanced activation between *CD44* and *VIM* significantly increased the M $\rightarrow$ E transition action while slightly decreased the E $\rightarrow$ M action, suggesting that stronger activation of M-state markers stabilizes the M state. Conversely, increased activation of E-state markers (e.g., *MUC1* and *CDH1*) raised the E $\rightarrow$ M transition action and lowered the M $\rightarrow$ E action, favoring MET and stabilizing the E state. Moreover, stronger mutual inhibition revealed that increased inhibition of E markers reduced the likelihood of MET, whereas enhanced inhibition of M markers made cells in the M state more prone to reverting to the E state. These sensitivity results are qualitatively consistent with known experimental observations⁶⁴.

To identify key genes driving the EMT, we conducted simulated knockout experiments. Our results show that knocking out *CD44* in M-state cells caused a reversion of cell fate back to the E state (Figure 6H, right), indicating that *CD44* is crucial for maintaining the M state. In contrast, enforcing low expression of *CDH1* in E-state cells promoted their transition to the M state (Figure 6H, left), consistent with the critical role of E-cadherin loss (encoded by *CDH1*) in EMT⁶⁵. Furthermore, perturbing the regulatory

parameters to assess the influence of *CD44* and *CDH1* on other genes revealed that changes in these two genes produce significant effects, underscoring their importance in EMT regulation. We analyzed gene expression trends along the MET transition pathway to determine the roles of different genes in the MET process. We found that the transition from the M state to the IM state is primarily driven by *MUC1*, while the expression of *VIM* (Vimentin) undergoes significant changes before cells reach the E state, consistent with its role as a downstream gene^{66,67} (Figure 6G).

To investigate cell transitions during EMT, we incorporated exogenous TGF- β into our model, reflecting its role in regulating gene expression. We treated the EMT process as a perturbation of the underlying dynamics trained on the MET process and tracked the EMT dynamics by fine-tuning the parameters of DynNet. TGF- β is present during the first 10 days of EMT but absent in the subsequent MET phase⁶³. It inhibits E-cadherin expression via degradation mechanisms involving *SMURF1* and *SMURF2*^{64,68}. To capture this, we added TGF- β as a regulatory node affecting *CDH1* and modelled its inhibitory effect as a function of its concentration⁶⁹. The degradation rate constant k was trained solely on the EMT process. We then compared the stable state distribution estimated by DynNet with real data (Figure 6D and 6E). The marginal distribution of the TGF- β -induced EMT process estimated by DynNet shows good agreement with the data at stable state distribution. This indicates that although the parameters of DynNet were trained on MET data, it can still successfully simulate the EMT process through fine-tuning with TGF- β . In addition, state transition probabilities were calculated (see Supplementary Information Section F.2). Our analysis revealed that the original E state (Day 0) comprises multiple sub-states, E1, E2 and E3, with E1 characterized by high expression of epithelial markers and E2/E3 by partial mesenchymal features. Notably, in the model retrained by adjusting only the degradation rate parameters, cells in the E1 state first switched to the E2 state before undergoing EMT (Figure S35). Transition probability calculations further supported a lower transition action for the E1-to-E2 pathway, indicating a higher likelihood for this route. Overall, these results suggest that our original dynamical model effectively captures the essential characteristics of the EMT dynamics. By training the regulatory parameters using

MET data and fine-tuning only the degradation rate parameters, we accurately described the dynamics of the EMT process, in line with the known role of TGF- β in modulating degradation rates, particularly for E-cadherin.

Collectively, these results demonstrate that DynNet effectively captures the essential characteristics of the EMT process by integrating MET training data, fine-tuning degradation rates, and incorporating TGF- β effects.

Discussion

Existing methods for inferring dynamics involve trade-offs between interpretability, data requirements, and scalability^{18,20,21}. DynNet overcomes these limitations by integrating mechanistic biophysical modeling with data-driven neural SDE frameworks. Unlike purely model-driven approaches constrained by oversimplified assumptions or black-box deep learning models that lack interpretability, DynNet explicitly parameterizes gene regulatory interactions using Hill functions while employing a dynamic noise estimator to capture the effects of stochasticity. By parameterizing causal regulatory mechanisms directly within the continuous drift function, it moves beyond traditional models that rely on post-hoc, local Jacobian matrix approximations, thereby granting the inferred dynamics robust global interpretability. Importantly, DynNet is fundamentally designed as a continuous dynamics modeling framework rather than a standalone Gene Regulatory Network (GRN) inference tool. This continuous-time integration inherently mitigates interval-induced spurious correlations that typically plague discrete-time models. Balancing prior biological information with data evidence, its integrated biophysical component captures both direct and indirect regulatory relationships within the gene network, facilitating dynamics inference from a minimal set of key genes directly in the original gene expression space.

On simulation datasets (MISA and branching process), we evaluate the capability of DynNet to infer stable state properties, predict cell fates, and track underlying dynamics, highlighting the value of integrating prior knowledge via constraints. On realistic hepatocyte differentiation dataset, DynNet reconstructs a dynamic landscape featuring two attractor states (MH and LB) and captures bifurcation

dynamics at the HE cluster, consistent with experimental observations. On EMT dataset, DynNet quantitatively characterizes the dynamics of EMT and MET processes, predicts transition pathways and identifies key genes as fate determinants, bridging molecular mechanisms with cellular decision-making. This capability allows for direct gene perturbation analysis and facilitates the identification of key genes and mechanistic rules driving cellular state changes. Ultimately, this framework presents a valuable tool for studying functions and mechanisms of complex biological systems, controlling cell fate transitions and proposing therapeutic strategies for disease^{28,70}.

To ensure the practical applicability and robustness of this mechanistic framework, several specific strategies and design choices have been incorporated into DynNet. First, to ensure computational feasibility in high-dimensional stochastic modeling, we adopted a diagonal diffusion matrix. Both theoretical and empirical validations confirm the robustness of this simplification: the continuous drift function effectively captures underlying gene-gene correlations, while the learned state-dependent noise dynamically compensates for omitted non-diagonal terms. Second, to facilitate the optimization process, we strategically integrated prior knowledge. While purely data-driven priors are susceptible to spurious correlations, our framework strictly utilizes them as optimization accelerators rather than validation metrics. To mitigate these inherent structural inaccuracies during *de novo* discovery, we employ an L2 soft-constraint, enabling continuous data dynamics to stably override false prior edges.

Nevertheless, there are several limitations related to structural assumptions and extreme computational scaling. First, while employing Hill functions significantly enhances mechanistic interpretability, it imposes a rigid structural assumption on the system's dynamics. This restricts the functional flexibility of the drift equation and constrains the optimization landscape. Second, the framework operates under the assumption that the biological system eventually converges to quasi-steady states, which limits its utility in analyzing highly oscillatory or cyclical processes, such as the cell cycle. Finally, despite our mathematical simplifications, the numerical integration of SDEs and the optimization of non-linear drift functions remain computationally demanding. Because this optimization is intensive, DynNet is

inherently constrained to targeted sets of core driver genes and cannot currently process genome-wide transcriptomic data now. Given these boundaries, DynNet is optimally positioned as a downstream analytical tool for deep mechanistic dissection rather than broad de novo network discovery.

To enhance DynNet's applicability, flexibility, and scalability, several strategies may be utilized. First, a gene encoding method may be adapted to analyze co-expression modules of high-dimensional gene data. Furthermore, a mixed modeling architecture that utilizes lightweight response functions for non-core gene modules can be used. To increase mechanistic modeling flexibility, it may be useful to incorporate a hybrid neural network architecture that combines the core mechanistic Hill function assumption with an assumption-free component (such as a general MLP), allowing DynNet to better describe complex system characteristics.

Methods

Neural network architecture

The proposed network structures are illustrated in Figure 1. Given that cell fate evolution can be expressed by SDEs

$$d\mathbf{x} = \mathbf{f}(\mathbf{x})dt + \boldsymbol{\sigma}(\mathbf{x})d\mathbf{W}, \quad (5)$$

the network structure was designed based on the principle of using nonlinear functions to model system dynamics and to account for fluctuations within the biological system.

In the proposed Neural SDE framework, the time variable t corresponds strictly to the experimental sampling time (i.e., the elapsed physical time between measurements) rather than an algorithmically inferred pseudotime. In biological developmental systems, cells sampled at a single experimental time point often exhibit significant stage heterogeneity, such as the coexistence of progenitor and differentiated cells. The SDE framework does not assume developmental synchrony within a given time point; rather, it explicitly models this within-timepoint heterogeneity as stochastic fluctuations, allowing individual cells

to progress at varying rates. Utilization of experimental real time preserves the temporal asynchrony necessary for estimating true kinetic quantities, such as transition rates and barrier-crossing timescales.

Velocity estimator

A neural network was developed to estimate $\mathbf{f}(\mathbf{x})$ in Eq.5. The input layer receives gene expression data, then transforms it through a hidden layer using non-linear activation functions inspired by Hill kinetics^{71,72}, which is a form widely applied in describing gene regulation in biological systems⁴³⁻⁴⁵. These transformations help the network capture gene regulatory effects in a biologically plausible manner.

The overall network structure is defined as follow,

$$\mathbf{f}(\mathbf{x}) = \mathbf{A}\mathbf{a}(\mathbf{x}) + \mathbf{B}\mathbf{r}(\mathbf{x}) - \mathbf{k} \circ \mathbf{x}, \quad (6)$$

where $\circ$ denotes the Hardmard product between vectors, and $\mathbf{k}$ is the vector of degradation rates. The regulatory coefficients ($\mathbf{A}, \mathbf{B}$) are the weight parameters of the output layer. These matrices are defined such that rows index the target gene i and columns index the source gene j . Therefore, the matrix element A_{ij} represents the activation strength of gene j on gene i , and B_{ij} represents the inhibition strength of gene j on gene i . To ensure biological interpretability, these matrices are maintained non-negative, allowing the model to decouple activating and inhibitory effects. The activation vector $\mathbf{a}(\mathbf{x})$ and repression vector $\mathbf{r}(\mathbf{x})$ are constructed by using the Hill function with the input state $\mathbf{x}$. Specifically, the j -th component of $\mathbf{a}(\mathbf{x})$ and $\mathbf{r}(\mathbf{x})$ is defined as:

$$a_j(x_j) = \frac{x_j^{n_h}}{S_j^{n_h} + x_j^{n_h}}, \quad r_j(x_j) = \frac{S_j^{n_h}}{S_j^{n_h} + x_j^{n_h}}, \quad (7)$$

where n_h denotes Hill coefficient (see Supplementary Information Section A.2). $\mathbf{S}$ is a vector representing the half-maximal effective concentration parameters (S_j for gene j). The vector $\mathbf{a}(\mathbf{x})$ is therefore composed of M component, $\mathbf{a}(\mathbf{x}) = [a_1(x_1), \dots, a_M(x_M)]^T$, and similarly for the repression

vector $\mathbf{r}(\mathbf{x})$. A neural network $\mathbf{f}_\theta$ was developed to estimate the drift term $\mathbf{f}(\mathbf{x})$ in Eq.5. The input layer receives gene expression data, then transforms it through a hidden layer using non-linear activation functions explicitly informed by Hill kinetics. These transformations help the network capture gene regulatory effects in a biologically plausible manner. The trainable parameter set of the network $\mathbf{f}_\theta$ is divided into two components: $\theta_1 = (\mathbf{A}, \mathbf{B})$ and $\theta_2 = (\mathbf{S})$. Specifically, $\mathbf{S}$ denotes the threshold vector associated with the Hill functions in the hidden layer. This vector modulates the sensitivity and steepness of the Hill activation functions, thereby enabling the network to model gene regulatory interactions. The parameters $(\mathbf{A}, \mathbf{B})$, serving as the weights of the output layer, make linear transformation on the embedded feature and generate the drift of SDE.

To ensure biological plausibility, we imposed constraints for all matrix elements $A_{ij}, B_{ij} \geq 0$, reflecting the activating or enhancing effects commonly observed in gene regulation processes, and $S_i > 0$, reflecting the half-saturation constant of gene expression is always a positive real number.

Noise estimator

Stochasticity in gene expression plays vital roles in many of the real biological processes, including differentiation and cell type transition^{73,74}. A Multi-Layer Perceptron (MLP) D_θ is implemented to approximate the diffusion matrix $\mathbf{D}(\mathbf{x})$, where $2\mathbf{D}(\mathbf{x}) = \boldsymbol{\sigma}(\mathbf{x})\boldsymbol{\sigma}(\mathbf{x})^T$ in Eq.5. Specifically, we assume $\mathbf{D}(\mathbf{x}) \in \mathbb{R}^{M \times M}$ is a diagonal matrix with diagonal elements $D_{ii}(\mathbf{x}) = d_i(\mathbf{x})$ and off-diagonal elements is zero. The MLP takes the gene expression profile of each cell as input and outputs the diffusion term for each gene at the cell's current state. This design enables adaptation to intricate patterns in noise generation, offering a biologically plausible representation of noise intensity across varying expression states.

Loss function and optimization strategy

Loss function design

When the noise level is low in Eq.5, the amplitude of the stochastic perturbation for the diffusion term has small Wasserstein p-norm. This phenomenon is called vanishing viscosity, and a rigorous description about it is in Supplementary Information Section A.3. In this way, we separate the estimation of stochastic dynamics Eq.5 into 2 steps. The two networks are optimised independently, first optimizing the parameters of $\mathbf{f}_\theta(\mathbf{x})$ and then optimizing the parameters of $\mathbf{D}_\vartheta(\mathbf{x})$.

For gene expression data spanning T time points, denoted as $\mathbf{X} = \{\mathbf{X}_1, \dots, \mathbf{X}_t, \dots, \mathbf{X}_T\}$, where $\mathbf{X}_t = \{\mathbf{x}_1, \dots, \mathbf{x}_{N_t}\}$ is the expression of all N_t cells at time t, the following three loss terms are considered when training the velocity estimator. First, a loss is introduced to constrain the system's behaviour at steady state. Assuming the final time point T represents steady state, the driving term in the stochastic differential equation should be close to zero. Therefore, the first loss term is defined as

$$\mathcal{L}_1 = \mathbb{E}_T[|\mathbf{f}(\mathbf{X}_t)|^2] = \frac{1}{N_T} \sum_{i=1}^{N_T} |\mathbf{f}(\mathbf{x}_i)|^2. \quad (8)$$

From the energy landscape theory (see Model-based landscape construction part in Methods and Supplementary Information Section B), the dynamic process can be modelled as multiple Gaussian peaks evolving their means and covariances, where the local properties ($\boldsymbol{\mu}_t$ and $\boldsymbol{\Sigma}_t$) are derived from the approximation of moment equations from Fokker-Planck equation. However, this local approximation inherently does not provide the global probability weights required to characterize the overall population distribution or track the temporal evolution of these state populations. The first term of Loss $\mathcal{L}_1$ can only ensure DynNet learn the local properties, but have no effect to match the probability weights of each state. To enable the velocity estimator to track total data dynamics across time points, a second loss term is defined by matching the predicted state weights at each time point. Specifically, for the experimental data, a K -component GMM is estimated at each time point t , obtaining the weights w_t^k for each Gaussian component, representing a stable state with K attractors. And we design a weight estimation

method to predict the weight $\widehat{w}_t^k$ from DynNet simulation data at each stable state. Utilizing these weights, the second loss function is defined as:

$$\mathcal{L}_2 = \frac{1}{T} \sum_{t=1}^T \mathbb{E}_t [|\widehat{\mathbf{w}}_t - \mathbf{w}_t|^2] = \frac{1}{TK} \sum_{t=1}^T \sum_{k=1}^K (\widehat{w}_t^k - w_t^k)^2, \quad (9)$$

where $\mathbf{w}_t$ is the weight vector estimated from the experimental data at time t , and $\widehat{\mathbf{w}}_t$ is the weight vector predicted by the model at time t . Specifically, the dynamic trajectory is simulated starting from the first data point using the ODE ($\frac{dx}{dt} = f_\theta(x)$), and sample points at the corresponding time points along this trajectory are taken as the estimated data $\widehat{X}_t$. The k -th component $\widehat{w}_t^k$ represent the weights of the k -th attractor, which is obtained by computing

$$\widehat{w}_t^k = \frac{1}{N_t} \sum_{i=1}^{N_t} R^k(\widehat{x}_i) P(C_k | \widehat{x}_i). \quad (10)$$

At a given estimated data point $\widehat{x}_i$, the relative likelihood $R^k(\widehat{x}_i)$ for the k -th Gaussian component is defined as the ratio of the probability density of $\widehat{x}_i$ under the k -th Gaussian distribution to the probability density at the mean of that distribution. The posterior probability $P(C_k | \widehat{x}_i)$ represents the probability that data point x_i belongs to the k -th Gaussian component (see Supplementary Information Section B). $\mathbf{w}_t$ is the vector of corresponding experimental data weights on each Gaussian component at time t .

Finally, to incorporate structural prior knowledge into the velocity estimator, we implemented both hard and soft constraint mechanisms based on a prior network matrix N , where $N_{ij} \in \{-1, 0, 1\}$ indicates the known regulatory relationship between nodes i and j . When using hard constraints, strict sparsity is enforced: parameters corresponding to non-existent edges ($N_{ij} = 0$) are rigidly fixed to 0 and excluded from the optimization process. Alternatively, when using soft constraints, we apply an L2 regularization

penalty to interactions that are absent from or contradict the prior network. Finally, a prior network constraint is considered for the velocity estimator. The constraint term $\mathcal{C}(\theta)$ for soft constraints is defined as

$$\mathcal{C}(\theta) = \sum_{i,j, N_{ij} \neq 1} A_{ij}^2 + \sum_{i,j, N_{ij} \neq -1} B_{ij}^2. \quad (11)$$

Combining these three loss terms, the total loss function for the velocity estimator is defined as:

$$\mathcal{L}_v = \mu_1 \mathcal{L}_1 + \mu_2 \mathcal{L}_2 + \lambda_1 \mathcal{C}(\theta). \quad (12)$$

During the alternating optimization of the output layer weights (**A**, **B**) and the hidden layer threshold vector **S**, different hyperparameters μ_1 and μ_2 are used for each optimization step. The entire optimization process is described in Algorithm 1. The alternating optimization stops when the change in the velocity field parameters $\theta_1 = (\mathbf{A}, \mathbf{B})$ between two consecutive iterations is below a predefined threshold ($\Delta\theta_1 < \xi$). The convergence criterion is calculated as:

$$\Delta\theta_1 = \sum |\theta_1^{(n)} - \theta_1^{(n-1)}|^2, \quad (13)$$

where n is the iteration index. The algorithm terminates if convergence is reached or the maximum number of iterations is exceeded. This framework aims to enable the learned parameters θ to capture the dynamic characteristics of the system's velocity field and stability.

Empirical observations indicate that the loss function $\mathcal{L}_2$ is mainly influenced by the parameter **S**, while the loss function $\mathcal{L}_1$ is influenced by both parameters **A**, **B** and **S**. Furthermore, the parameter **S** is not included in the constraint term. These findings have implications for the selection of hyperparameters in the algorithm.

Algorithm 1: Optimization Algorithm for Parameters

Input: Data $\mathcal{D} = \{\mathbf{X}_1, \dots, \mathbf{X}_T\}$, learning rates η_1, η_2 , number of epochs E_1, E_2 , velocity network $f_\theta(\mathbf{x})$ with parameters $\theta_1 = (\mathbf{A}, \mathbf{B})$ and $\theta_2 = \mathbf{S}$, threshold ε , hyperparameters μ_1, μ_2, λ_1 , maximum iterations N_{iter}

```

1  Initialize: Randomly initialize  $\theta_1$  and  $\theta_2$ .
2  Set iteration counter  $n \leftarrow 0$ .
3  Set parameter change  $\Delta\theta_1 \leftarrow \infty$ .
4  While  $\Delta\theta_1 \geq \xi$  and  $n < N_{\text{iter}}$ 
5       $\theta_1^{(n)} \leftarrow \theta_1$ .
6      For  $e = 1, 2, \dots, E_1$ 
7          Compute loss  $\mathcal{L}_v \leftarrow \mu_1 \mathcal{L}_1 + \lambda_1 \mathcal{C}(\theta)$ .
8          Update  $\theta_1 \leftarrow \theta_1 - \eta_1 \nabla_{\theta_1} \mathcal{L}_v$ .
9      End
10     For  $e = 1, 2, \dots, E_2$ 
11         For  $t = 1, 2, \dots, T$ 
12             Estimate GMM weights  $w_t$  from experimental data  $X_t$ .
13             Simulate data  $\hat{\mathbf{X}}_t$  using  $f_\theta(\mathbf{x})$ .
14             Estimate GMM weights  $\hat{w}_t$  from simulated data  $\hat{X}_t$ .
15             Compute loss  $\mathcal{L}_2 \leftarrow \mathcal{L}_2 + \mathbb{E}_t[|\hat{\mathbf{w}}_t - \mathbf{w}_t|^2]$ .
16         End
17         Compute loss  $\mathcal{L}_1 \leftarrow \mathbb{E}_T[|f_\theta(X_t)|^2]$ 
18          $\mathcal{L}_S \leftarrow \mu_1 \mathcal{L}_1 + \frac{\mu_2}{T} \mathcal{L}_2$ .
19         Update  $\theta_2 \leftarrow \theta_2 - \eta_2 \nabla_{\theta_2} \mathcal{L}_S$ .
20     End
21      $n \leftarrow n + 1$ 
22     Compute  $\Delta\theta_1 = \sum |\theta_1 - \theta_1^{(n)}|^2$ .
23 End

```

Return: Optimised parameters θ_1 and θ_2 .

Next, with the parameters of the velocity estimator θ fixed, the parameters of the noise network ϑ are optimised. During this phase, the parameters θ_1 and θ_2 remain fixed while only the noise network parameters ϑ are optimised to calibrate the noise characteristics of the system. The noise parameters are

optimized by minimizing the discrepancy between the observed probability distribution $p_{j,t}$ and the model-predicted distribution $\hat{p}_{j,t}$ for each gene j at time t . We employ the Kullback-Leibler (KL) divergence to quantify this difference. To obtain the marginal distributions required for this calculation, we utilize distinct estimation strategies for the experimental and DynNet simulation data. For the experimental data, the overall joint distribution has been estimated using a GMM in the velocity estimation step. The marginal distribution $p_{j,t}$ for the j th gene is analytically derived. The predicted distribution $\hat{p}_{j,t}$ of DynNet simulation data is derived from data points generated through dynamic simulations by DynNet, with its marginal density estimated using the non-parametric method, Kernel Density Estimation (KDE). The goal is to minimize this discrepancy, thereby optimizing the noise parameters to better fit the stochastic characteristics in the data. The resulting loss function for the noise network is defined as

$$\mathcal{L}_D = \frac{1}{T} \sum_{t=1}^T \mathbb{E}_t [\text{KL}(\hat{p}_{j,t}|p_{j,t})] + \lambda_2 \mathcal{R}_2(\vartheta), \quad (14)$$

where $\mathcal{R}(\vartheta)$ is the regularization term. The specific optimization process is detailed in Algorithm 2.

Algorithm 2: Optimization Algorithm for Noise Network Parameters ϑ

Input: Training data $\mathcal{D} = \{\mathbf{X}_1, \dots, \mathbf{X}_T\}$, learning rate η_3 , number of epochs E_3 , noise network $D_\vartheta(\mathbf{x})$ with parameters ϑ , fixed velocity estimator $f_\vartheta(\mathbf{x})$,

- 1 **Initialize:** Randomly initialize ϑ .
- 2 **For** $e = 1, 2, \dots, E_3$
- 3 **For** $t = 1, 2, \dots, T$
- 4 Estimate marginal distribution p_t from experimental data X_t .
- 5 Simulate data $\hat{\mathbf{X}}_t$ using $f_\vartheta(\mathbf{x})$ and $D_\vartheta(\mathbf{x})$.
- 6 Estimate marginal distribution $\hat{p}_t$ from simulated data $\hat{\mathbf{X}}_t$.
- 7 Compute loss for all genes $\mathcal{L}_D \leftarrow \mathcal{L}_D + \sum_{j=1}^M \text{KL}(\hat{p}_{j,t}|p_{j,t})$.
- 8 **End**
- 9 $\mathcal{L}_D \leftarrow \mathcal{L}_D + \lambda_2 \mathcal{R}(\vartheta)$.
- 10 Update parameters: $\vartheta \leftarrow \vartheta - \eta_3 \nabla_\vartheta \mathcal{L}_D$.

11 End

Return: Optimised parameters ϑ .

A detailed analysis of the time complexity is essential to demonstrate the scalability of the DynNet model on large single-cell datasets. Let M be the number of genes, N be the average number of cells at each time point, and L be the number of time steps. The total complexity is dominated by the optimization phases within Algorithm 1 and Algorithm 2.

For the Algorithm 1, the complexity is primarily driven by two steps: the network optimization, which involves a dominant $\mathcal{O}(M^2)$ operation, and the numerical solving of the ODE via the Euler method, which has a complexity of $\mathcal{O}(NM^2L)$. The overall time complexity for Algorithm 1 is thus approximated as $\mathcal{O}(NM^2L)$. Similarly, the complexity of Algorithm 2, which involves SDE simulation using the Euler method, is also $\mathcal{O}(NM^2L)$.

GRN Reconstruction

Although DynNet is fundamentally designed for continuous dynamics modeling rather than standalone GRN inference, the constraint method provides utility of GRN reconstruction under purely data-driven settings. Specifically, when the model is optimized using soft constraints without any external prior knowledge, this procedure enables the *de novo* generation of a reliable reference GRN. This extracted topology can subsequently be utilized as a structural constraint in subsequent training phases to further stabilize and guide the inference of the continuous vector field. To discretize the continuous learned weights into a final sparse network topology without relying on arbitrary, hard-coded cutoffs, we implemented a data-driven thresholding strategy based on GMM (see Supplementary Information Section A.5). Under L2 regularization, the distribution of these normalized weights typically exhibits a dense component near zero (representing regularized, non-essential edges) and a long tail (representing data-supported interactions). We fitted a two-component GMM to this distribution. The component with the lower mean was designated as the background noise model, parameterized by a mean (μ_{noise}) and a

standard deviation (σ_{noise}). The specific cutoff for retaining a structural edge was defined as the 99.7% confidence upper bound of this noise distribution ($\mu_{\text{noise}} + 3\sigma_{\text{noise}}$). Edges with absolute weights exceeding this threshold were retained in the final discrete GRN.

Energy landscape construction

Understanding the stable states of a system is crucial in the study of cell fate decision. The concept of an energy landscape serves as a valuable tool for this, likening the system's state space to a topographical map where low potential energy regions correspond to more stable states, i.e., states where gene expression is more likely to reside. Different approaches have been developed to quantify the energy landscape for gene networks^{41,44,54}. Specifically, the energy landscape can be inferred from the stable state probability distribution $P_{\text{ss}}(\mathbf{x})$ using the formula

$$U(\mathbf{x}) = -\ln P_{\text{ss}}(\mathbf{x}). \quad (15)$$

We employ two distinct approaches for constructing energy landscapes depending on information availability: when a stochastic dynamical model (Eq. 5) is established, we quantify the landscape from Gaussian approximation approach; conversely, in the absence of an explicit model, we estimate the landscape directly from the observed data under a quasi-steady-state approximation.

Model-based landscape construction

If the system's dynamics can be described by Eq.5, the corresponding Fokker-Planck equation can be used to estimate the energy landscape. This equation describes the temporal evolution of the probability distribution $P(\mathbf{x})$, and it combines deterministic motion governed by the drift force $\mathbf{f}(\mathbf{x})$ with stochastic perturbations determined by the diffusion matrix $\mathbf{D}(\mathbf{x})$

$$\partial_t P(\mathbf{x}) = -\nabla \cdot (\mathbf{f}(\mathbf{x})P(\mathbf{x})) + \nabla \cdot (\mathbf{D}(\mathbf{x})\nabla P(\mathbf{x})), \quad (16)$$

When the diffusion is small, the evolution of the probability distribution can be approximated using moment equations^{41,44}. For a Gaussian distribution, its probability distribution is fully determined by its mean $\boldsymbol{\mu}(t)$ and covariance $\boldsymbol{\Sigma}(t)$. These moments evolve according to the following equations

$$\frac{d}{dt}\boldsymbol{\mu}(t) = \mathbf{f}(\boldsymbol{\mu}(t)) \quad (17)$$

$$\frac{d}{dt}\boldsymbol{\Sigma}(t) = \mathbf{J}(\boldsymbol{\mu})\boldsymbol{\Sigma}(t) + \boldsymbol{\Sigma}(t)\mathbf{J}^T(\boldsymbol{\mu}) + 2\mathbf{D}(\boldsymbol{\mu}), \quad (18)$$

where $\mathbf{J}$ is the Jacobian matrix of the drift force $\mathbf{f}(\mathbf{x})$. As time goes to infinity, the system reaches a stable state, and the mean $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ at a specific attractor can be obtained by solving the above Eq. (6). Then, the probability density distribution for an M-dimensional system at one stable state can be expressed as

$$P(\mathbf{x}) = \frac{1}{(2\pi)^{\frac{M}{2}}|\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right). \quad (19)$$

When the system has multiple attractors, the stable state probability distribution can be estimated as a mixture model of them. By solving Lyapunov equation (6) at each attractor $\{\boldsymbol{\mu}_i\}_{i=1}^N$, we get covariances $\{\boldsymbol{\Sigma}_i\}_{i=1}^N$, and the probability density function of the system can be computed as

$$P_{ss}(\mathbf{x}) = \sum_{i=1}^N \frac{w_i}{(2\pi)^{\frac{M}{2}}|\boldsymbol{\Sigma}_i|^{\frac{1}{2}}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu}_i)^T \boldsymbol{\Sigma}_i^{-1}(\mathbf{x} - \boldsymbol{\mu}_i)\right). \quad (20)$$

Here w_i is weight of each attractor. For detailed computation, one may refer to our previous works^{41,45}.

Data-driven landscape construction

In contrast to the model-based approach discussed previously, which relies on knowing the system's dynamical equations, the data-driven approach is employed when these equations are not available. In

such cases, the system's steady state probability distribution can be estimated directly from experimental data under the assumption that the system is near a stable state after a period of evolution. A commonly used method is to approximate this distribution using a GMM

$$P_{ss}(\mathbf{x}) = \sum_{k=1}^K w_k \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k), \quad (21)$$

where K represents the number of Gaussian components.

Since the true number of stable states is often unknown, we utilize the Bayesian Information Criterion (BIC) to address this. Similar to the model-based landscape, each Gaussian component is described by its weight w_k , mean $\boldsymbol{\mu}_k$, and covariance matrix $\boldsymbol{\Sigma}_k$. By fitting the experimental data, these quantities can be obtained, and further used to approximate the steady state probability distribution and the energy landscape.

Dimension reduction for landscape

Gene regulatory networks often involve a large number of genes, resulting in a high-dimensional state space that can be challenging to interpret directly. To facilitate visualization and analysis of the energy landscape, it is often necessary to project it onto a lower-dimensional subspace. We followed the method in our recent work to reduce the high-dimension landscape⁴¹. A common and effective technique for this dimensionality reduction is based on the principal components of the system's variability, which can be extracted from the covariance matrix of the data.

Specifically, a projection matrix $\mathbf{V} \in R^{M \times C}$ is constructed, where M is the original high dimensionality and $C < M$ is the desired lower dimensionality. The columns of $\mathbf{V}$ are formed by the C eigenvectors corresponding to the first C largest eigenvalues of the covariance matrix $\boldsymbol{\Sigma}$ of the system's data. These eigenvectors represent the directions of maximum variance in the data.

Given the data X_j for the j -th stable state in the original high-dimensional space, the projected data Z_j in the low-dimensional space is obtained by

$$\mathbf{Z}_j = \mathbf{V}^T \mathbf{X}_j. \quad (22)$$

Similarly, if μ_j and Σ_j are the mean and covariance matrix of the j -th stable state in the original space, their counterparts in the projected C -dimensional space, μ_j^z and Σ_j^z , are given by

$$\mu_j^z = \mathbf{V}^T \mu_j, \quad \Sigma_j^z = \mathbf{V}^T \Sigma_j \mathbf{V}. \quad (23)$$

The probability density distribution of the j -th stable state in this projected C -dimensional space can then be represented by a C -dimensional multivariate normal distribution

$$p_j^z(\mathbf{z}) = \frac{1}{(2\pi)^{C/2} |\Sigma_j^z|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{z} - \mu_j^z)^T (\Sigma_j^z)^{-1} (\mathbf{z} - \mu_j^z)\right). \quad (24)$$

Through this projection method, the complex, high-dimensional energy landscape can be effectively reduced to a few key dimensions, allowing for easier visualization of the stable states (attractors) and the potential barriers or transition pathways between them. This simplified representation facilitates a more intuitive understanding of the system's dynamics and stability properties.

Evaluation

To systematically characterize the properties, dynamics, and predictive capability of the system under investigation, we employ a comprehensive set of quantitative metrics and computational analyses. These evaluations target five principal aspects: transition dynamics, classification accuracy, distributional similarity, temporal characteristics, and vector field reconstruction.

Minimum Action Path (MAP). The most probable transition trajectories between stable states are identified by minimizing the Freidlin-Wentzell action functional, $S_T(x)$, using the Adaptive Minimum Action Method (aMAM)⁷⁵. The action functional quantifies the deviation from deterministic dynamics

$$S_T(\mathbf{x}) = \frac{1}{2} \int_0^T \|\dot{\mathbf{x}}(t) - \mathbf{f}(\mathbf{x}(t))\|_2^2 dt \quad (25)$$

where $f(x(t))$ is the drift term. Minimizing $S_T(x)$ yields the most probable transition path for state switching in the presence of noise.

Classification Boundaries and F1 Score Comparison. To assess the system's state predictability and the performance of potential classifiers, attractor basins and their separating decision boundaries $\mathcal{B}$ are defined. The effectiveness of assigning states x to their correct final attractor $\phi(x)$ is measured using F1 scores, calculated from True Positives (TP), False Positives (FP), and False Negatives (FN). The class-specific F1 score is

$$F1_k = \frac{2TP_k}{2TP_k + FP_k + FN_k} \quad (26)$$

Maximum Mean Discrepancy (MMD). This non-parametric metric quantifies the dissimilarity between the probability distributions of two sample sets, X and Y , by comparing their embeddings in a reproducing kernel Hilbert space $\mathcal{H}$. The squared MMD is defined as the squared distance between the mean embeddings

$$\text{MMD}^2(X, Y) = \|E_{x \sim P_X}[\kappa(x, \cdot)] - E_{y \sim P_Y}[\kappa(y, \cdot)]\|_{\mathcal{H}}^2 \quad (27)$$

where κ is the chosen kernel (e.g., RBF kernel $k(x, y) = \exp(-|x - y|^2/2\sigma^2)$). MMD provides a rigorous way to determine if observed differences between datasets are statistically significant, useful for comparing experimental conditions or model simulations.

Velocity Field Estimation. The underlying system dynamics are reconstructed from discrete observations (e.g., single-cell measurements in an embedding space) by estimating a continuous velocity vector field.

First Passage Time (FPT) Analysis. FPT measures the time required for a stochastic process X_t governed by an SDE to first reach a predefined target state or region, starting from $X_0 = x_0$. The FPT between different cell states can be estimated through multiple simulations.

Statistics and Reproducibility

To evaluate the robustness of the DynNet framework, all computational findings were validated through multiple independent simulation runs using different random seeds. For the MISA and branch synthetic datasets, individual cells were simulated independently; the robustness of these simulations was verified across various random seeds, with the exact number of simulated cells detailed in respective figure legends. No statistical method was used to predetermine sample size. Rather, sample sizes for stochastic simulations were chosen based on standard practices in computational systems biology to ensure the stability of the inferred dynamic distributions and sufficient statistical power for transition analysis. Specifically, for the hepatocyte dataset, FPT estimations involved 100 independent stochastic simulations per temporal resolution condition to properly elucidate differentiation kinetics. For the simulated gene knockouts, 50 independent cell trajectories were simulated for each knockout and wild-type control group, with statistical significance between transition delays determined using a two-sided Wilcoxon rank-sum test. Similarly, for the EMT dataset, the perturbation-induced changes in the minimum action path (ΔS) were calculated as the average over 50 independent simulation replicates. All successfully completed simulation runs were included in the reported results, and no data were excluded from the analyses. While inherent stochasticity is introduced via the simulation algorithms, the experiments were not randomized in terms of group allocation. Furthermore, the investigators were not blinded to allocation during experiments and outcome assessment, as the group assignments—such as specific *in silico* gene knockouts—were mathematically predefined by the computational models.

Datasets and prior knowledge

This study utilizes both synthetic and experimental single-cell RNA sequencing (scRNA-seq) data. Synthetic data, generated using the MISA and BoolODE models, was used to evaluate the method's performance⁵⁶. These data were split into training and testing sets (see Supplementary Information Section E.1 and E.2). The MISA model simulates multi-stable gene regulatory networks, while the BoolODE model combines Boolean logic with ordinary differential equations. Experimental scRNA-seq data from publicly available sources, specifically an EMT dataset⁶³ and a hepatocyte differentiation dataset⁴⁹, were pre-processed using *Scanpy*.

Prior knowledge about gene regulatory relationships was integrated into the model. This information, extracted from databases like *STRING* and *EVEX*, was used to construct a prior gene regulatory network. In addition, we also consider using correlation as an estimation of the gene regulatory network. Both hard and soft constraints are employed to guide the learning process. Hard constraints enforce consistency with highly reliable prior knowledge, while soft constraints allow the model to adapt when prior information is less certain or incomplete. These constraints help to improve the biological interpretability of the model by encouraging it to learn regulatory patterns consistent with existing knowledge.

Data Availability

The epithelial-to-mesenchymal transition (EMT) single-cell data used in this study are available in the Cytobank Stanford database under experiment ID 26555 and from the Source Data file associated with the original publication at <https://www.nature.com/articles/s41467-019-13441-6>. The hepatocyte development single-cell RNA-seq data used in this study are available in the Gene Expression Omnibus (GEO) database under accession code GSE96981. The synthetic datasets generated and analyzed in this study have been deposited in the GitHub repository at <https://github.com/cetusdou/DynNet/tree/main/data>. Source data are provided with this paper. No restricted-access datasets were generated or analyzed in this study.

Code Availability

The source code of DynNet to reproduce the results of this study is available on GitHub

<https://github.com/cetusdou/DynNet>.

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Author Contributions

C.L. and J.D. conceived and designed the research, J.D. performed the research, C.L., J.D., W.L., F.C., and Q.N. analyzed the data, and J.D., Q.N., and C.L. wrote the manuscript.

Competing interests

The authors declare no competing interests.

Figure Captions

Figure 1: Overview of the DynNet framework.

A, Single-cell sequencing technology collects cells from multiple time points of a biological process and performs sequencing. After converting the sequencing data into a count matrix and performing preprocessing, the data is input into DynNet. **B**, By utilizing correlations and databases, a prior directed gene expression network can be obtained to constrain DynNet, where $\neg$ and $\rightarrow$ in the network represent inhibitory and activating effects. **C**, The DynNet framework includes two sub-networks. Left is the Velocity Estimation Network $\mathbf{f}_\theta(\mathbf{x})$, which extracts mechanistic features using Hill function kinetics (parameterized by threshold S and coefficient n_h) and combines them through linear regulatory strength (w , which are decomposed into activation matrix A and inhibition matrix B) to predict gene expression velocity. Right is the Noise Estimation Network $\mathbf{D}_\theta(\mathbf{x})$, which estimates the diffusion term. Both networks take gene expression data as input. The velocity network outputs the velocity (drift) values, while the noise network outputs the diffusion matrix (constrained as diagonal) representing stochastic fluctuations for each data point. **D**, A GMM is used to estimate the data characteristics of scRNA-seq, which serves as a prior for calculating the loss of the neural network. **E**, DynNet training involves alternating training for the velocity estimator's parameters (thresholds and regulation parameters), followed by independent training of the noise estimator. Key loss functions include drift loss, weight loss, and KL divergence. Hyperparameters for combining these losses are determined during pre-training. **F**, The dynamics modelling of DynNet is achieved by constructing and solving moment equations to obtain the energy landscape, which quantifies the global dynamical behaviour of the system. **G**, Downstream analysis tasks of DynNet include generating the energy landscape, analysing stable state transition paths, predicting bifurcation points of cell fate, and analysing potential weighted regulatory networks. **H**, DynNet's capabilities are benchmarked against four commonly used dynamic analysis methods.

Figure 2: DynNet accurately recovers multi-stable dynamics of MISA model.

A, Network of a MISA system, where two genes mutually inhibit each other (blue $\rightarrow$) and exhibit self-activation (red $\rightarrow$). **B**, MISA identifies three distinct stable states (S1: exclusive expression of gene X1, S2: exclusive expression of gene X2, S3: median expression of both genes) from scatter plot data whose expression levels are shown in the table (see Supplementary Information Section E.1). **C**, DynNet accurately estimates cluster centers and covariances, demonstrating close alignment between the model-inferred and data-driven cluster centers. The estimated covariances effectively encompass the data distribution, validating the precision of the DynNet. **D**, The marginal distributions of gene expression levels for gene X1 and X2, as estimated by DynNet, show correspondence with the empirical data. **E**, The landscape and velocity field visualization inferred by DynNet reveal the topography of the energy landscape and its associated flow dynamics. The attractors in the landscape correspond to the stable states estimated by the MISA model, while the velocity field highlights the global dynamics of transitions from unstable states toward these attractors. **F**, Decision boundaries inferred from both empirical data and DynNet (highlighted by three distinct colors) clarify the partitioning of phase space among the three stable states. The consistency between the boundaries derived from the model and the simulation data suggests that DynNet's potential for predicting unobserved cellular dynamics. This synthetic dataset includes 10000 cells and 5 timepoints. **G**, Comparison of the MSE for the velocity field estimation between Neural ODE and DynNet. Each method is trained with 2, 3 and 5 time points, individually. **H**, The velocity fields of benchmark (from ground truth data), Neural ODE (trained with 5 data points) and DynNet (trained with 5 data points). Compared with Neural ODE, DynNet provides better estimation of the ridge (red circle) and saddle point (blue circle).

Figure 3: DynNet reconstructs dynamics of developmental branching process.

A, The GRN (7 gene network) with red arrows indicating activating regulation and blue arrows representing inhibitory regulation, reveals the underlying regulatory mechanisms of BoolODE that govern

gene expression dynamics. **B**, PCA visualization of the system's trajectories shows the bifurcation process in the simulation data, where each time point is colored according to its progression in the dynamical process, and developmental trajectories of 1000 independent simulated cells are mapped onto a low-dimensional manifold. The differentiation process starts from initial state (I) and two distinct stable states (D1 and D2) are marked by separate colors with darker shades representing later time points. **C**, Violin plots compare gene expression levels between the two stable states, with the x-axis representing different genes and the y-axis indicating expression levels. Each plot illustrates the distribution of expression values for a specific gene in the respective stable states (left: D1, right: D2). **D**, Histogram shows the alignments between the estimated marginal distribution of each gene with data. **E**, Gene expression velocity field plot depicts the temporal trends of gene expression changes, with arrow directions signifying trajectories largely consistent with the developmental paths in **B** and illustrating the system's evolution toward the stable states. **F**, Gene expression trajectories over time illustrate the dynamic expression patterns of genes G6 and G7 during the cell fate decision process, emphasizing the bifurcation behaviour of gene expression and its association with cell fate determination. **G**, The activation and inhibition parameter matrices quantify regulatory interaction strengths between genes under soft constraints, with results largely consistent with prior knowledge. Higher values indicate stronger effects, and inconsistent entries are highlighted by dashed outlines. **H**, Comparison of training loss curves in DynNet shows faster convergence under hard constraints, but soft constraints eventually achieve a similarly optimal result.

Figure 4: DynNet reconstructs hepatocyte differentiation and regulatory dynamics.

A, Schematic trajectory showing iPSCs sequentially progressing through DE, HE, IH, early/late LB, and MH states, resembling a potential energy descent. **B**, The reference GRN is depicted as a circular diagram, where each sector represents a key gene involved in hepatocyte differentiation—red arrows indicate activation interactions while blue arrows denote inhibitory interactions. The GRN is primarily derived from the dataset (see Supplementary Information Section F.1). **C**, Single-cell transcriptomic data

are visualized by projecting onto the first two principal components (PC1 and PC2). Cell states are color-coded and a black line traces the overall differentiation pathway. **D**, Density plots display the expression distributions of marker genes across various cell states, with distinct colors differentiating MH, LB, and intermediate states. **E**, A PCA-based dynamic landscape presents a stability map where arrows indicate transition trajectories. **F**, The state transition network is represented as a directed graph, with nodes corresponding to cell states and edges weighted by transition probabilities and durations, delineating the dynamics of state changes. **G**, Boxplots of first passage times (FPT) compare transition durations from DE to MH and DE to LB states under different temporal resolutions ($dt = 0.01$ and 0.1). **H**, Scatter plots reveal shifts in cell states following the knockout of latent regulatory genes (e.g., *TOB1*, *RANBP1*, *SRGAP2*), with altered state distributions highlighted. **I**, Boxplots further illustrate delays in transition times toward MH and LB states after simulated gene knockouts, underscoring the regulatory impact on differentiation timing. For all boxplots in **G** and **I**, center lines indicate medians; box bounds represent 25th/75th percentiles; whiskers extend to $1.5 \times$ IQR; points denote outliers. Statistical significance was determined using two-sided Wilcoxon rank-sum tests ($*p < 0.05$, $**p < 0.01$, $***p < 0.001$).

Figure 5: DynNet captures the transition dynamics in EMT from single-cell data.

A, Model-based landscape is generated from experimental data, delineating four distinct stable states: E, M, pEMT, IM. The landscape is reconstructed based on state probabilities observed in the dataset. **B**, One-dimensional projections of initial (day 10) and stable states (day 20) of EMT process illustrate the dynamic transition process, with landscape transformations elucidating the underlying cause of state evolution. **C**, The estimated landscape by DynNet, aligned with experimental data points, validates the model's accuracy in capturing dynamic features. The close correspondence between the two suggests that DynNet effectively reconstructs the system's stability properties. **D**, Comparison of model-estimated and experimental marginal distributions of stable states highlights the model's precision in reproducing stable state properties. Statistical metrics confirm the consistency between estimated and observed distributions. **E**, Estimated expression profiles of EMT core gene pairs reveal their regulatory relationships in stable

states, suggesting coordinated expression patterns that define phenotypic stability. **F**, Gene expression velocity field illustrates the directed dynamics toward stable states during MET, capturing the cell fate transitions. Arrow directions indicate estimated expression trajectories. **G**, Weighted gene regulatory network inferred by DynNet, with edge widths reflecting the potential strength of regulatory interactions, and nodes representing core genes selected.

Figure 6: DynNet predicts possible transition paths and identifies critical genes for EMT.

A, Possible direct transition paths between the E state and the M state identified using the minimum action path method (Red: M to E; White: E to M). **B**, Transition paths between the E state and the M state through intermediate states, illustrate the process of cell transition from the E state to the M state, including intermediate states (IM and pEMT). **C**, State transition network diagram depicts all transition pathways among stable states. Arrow thickness represents transition probabilities, while labelled values indicate the action associated with each transition path. **D**, Comparison of model-estimated and experimental marginal distributions of stable states in the TGF- β -induced EMT process, assessing the model's accuracy in replicating stable state properties. **E**, DynNet's estimated EMT landscape overlaid with experimental data points, validating the model's ability to reconstruct state transitions and dynamic stability. **F**, Parameter sensitivity analysis, showing the impact of changes in model weight parameters a and b on the transition action. The figure quantifies the effects of perturbations in parameters a and b on the transition actions from the E state to the M state ($\Delta S_{E \rightarrow M}$) and from the M state to the E state ($\Delta S_{M \rightarrow E}$), revealing parameter-dependent shifts in transition dynamics. **G**, Temporal changes in gene expression profiles during the transition from the M state to the E state via intermediate states, capturing key regulatory shifts along the trajectory. **H**, Results of simulated gene knockout experiments, showing the impact of knocking out *CDH1* and *VIM* genes on cell state transitions. The red areas in the figure indicate the states reached after knockout, while the grey areas represent the original experiment data.

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EDITORIAL SUMMARY

Dou and colleagues present DynNet, a biophysically informed deep learning framework that reconstructs continuous cellular dynamics from time-resolved single-cell data. DynNet maps cell fate transitions and offers insights into complex biological processes.

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